NAME: ______________________________ PERIOD: ________

Which Summary Tells the Story?

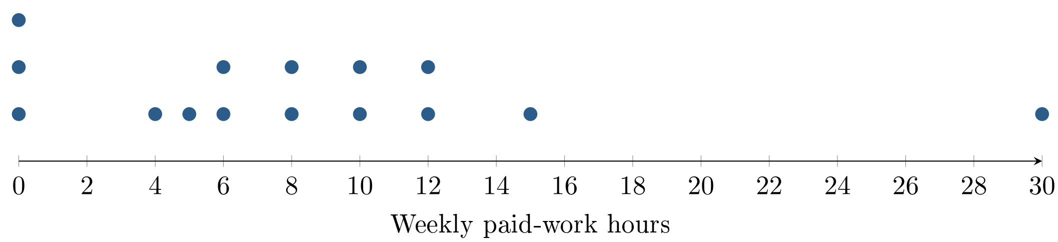
AP Statistics · Unit 1 · Topic 1.7

[STAR] THE PROBLEM

Suppose a principal asks, "How much paid work do students do in a week?" Fifteen students report these weekly hours: 0, 0, 0, 4, 5, 6, 6, 8, 8, 10, 10, 12, 12, 15, 30.

Kai: "$\bar{x} = 8.4$, $s_x \approx 7.5$, and IQR $= 8$. Report the mean and SD because they use every observation."

Rosa: "Median $= 8$, $Q_1 = 4.5$, $Q_3 = 11$, and IQR $= 6.5$. Report the median and IQR because 30 looks unusual."



FIRST TAKE — one sentence before you work the parts

Whose report would you send, Kai's or Rosa's? Cite one number or dotplot feature.

[BOOK] CENTER, SPREAD, AND OUTLIERS (keep on the page)

The median is the middle ordered value (average the two middle values for an even count). For this sheet, find Q_1 and Q_3 as the medians of the lower and upper halves, excluding the overall median when n is odd. IQR $= Q_3 - Q_1$. Outlier fences are $Q_1 - 1.5\text{IQR}$ and $Q_3 + 1.5\text{IQR}$. Mean and SD are nonresistant: extremes affect them more. Median and IQR are resistant. Use the distribution's shape to choose a pair.

Work (a)–(c).

DOK 1 **(a)** State the median and maximum weekly paid-work hours.

DOK 2 **(b)** Using the printed quartile rule, find Q_1, Q_3, the IQR, and the upper 1.5-IQR fence. Is 30 beyond it? Use Kai's mean and SD as given.

DOK 3 **(c)** Whose IQR follows the printed rule, and how did the other result arise? Choose a corrected summary pair for the principal. Use a dotplot feature to explain why that pair describes typical work hours better than the alternative.

Frame: __________'s IQR follows the printed rule; the other calculation includes ____________________.

Frame: I would report ________________ because the dotplot shows _________________________; the alternative could make the principal believe ____________________________.

Turn this sheet in whenever you finish — bonus credit. Part (c) is scored E / P / I.